

System Design Adherence Audit

Project: atmata-ui-only (InsForge ERP backend + Next.js UI)

Date: 30 August 2026 · Method: CEO brief + System agent + Data agent

Overall verdict: Partial adherence. The data layer largely matches the day-one design. The system shape only partly matches, because the live stack is Vercel plus InsForge, not one vertical server.

Intent of the design: one application, one SQL database, and one file store. Each company has few users and many writes. Screen loads must be fast. Roles control access. The server is stateless from day one. Later you can add a load balancer, a cache, and database replicas without a redesign.

1. CEO summary

Cause of the gap: the design describes one box (one app server, vertical scale). The product runs as Next.js on Vercel plus InsForge Postgres, Auth, Storage, and edge functions — a managed multi-instance cloud layout with a shared SQL database and company isolation.

Layer	Verdict	Note
Data layer	Largely adheres	InsForge Postgres ERP with RLS, posting, reports, file links
System components	Partially adheres	SQL and no Redis OK; one-server / vertical scale FAIL
Overall	Partial adherence	Close on data and later readiness; open on topology wording

2. Stack — four places

Place	Stack
Client	Next.js 16 + React 19 + Tailwind 4 + next-intl + @insforge/sdk (browser)
Server	Next Server Components / Actions on Vercel + InsForge edge functions
Database	InsForge PostgreSQL (project erp-backend-v1, shared + company_id RLS)
Storage	InsForge buckets documents and imports

3. System components checklist

Agent: System stack audit. Evidence from package.json, vercel.json, migrations, InsForge config, and auth/session code.

Rule	Status	Stack / note
Monolithic one deploy unit	PARTIAL	One UI repo; BaaS + edge functions are separate deploy units
Vertical scaling / one server	FAIL	Vercel serverless + managed InsForge (platform scales instances)
SQL only for core data	PASS	PostgreSQL; no Mongo/Redis for core
No cache in front of DB	PASS	No Redis/Upstash; React cache() = request dedupe only
One server topology	FAIL	Many Vercel instances + InsForge API + edge functions
One DB per company	PASS*	Shared Postgres + RLS (my_company_id), not one instance per company
1–30 users per company	UNKNOWN	No seat cap in schema or product docs

Rule	Status	Stack / note
Write-heavy workload	PARTIAL	Write RPCs + write_commands; lists still bulk-read
Fast loading lists/forms	PARTIAL	SSR yes; allPages can load full tenant tables
Upload delay permitted	PASS	Browser InsForge Storage; OCR jobs async
Stateless server	PARTIAL	Cookie/JWT OK; in-memory rate-limit Map is not
Load-balancer friendly	PARTIAL	Auth OK; process-local throttle breaks clean LB
Cache friendly (later)	PASS	No sticky sessions for auth; revalidatePath after writes
DB-replica friendly (later)	PARTIAL	Single write path via RPCs; no replica routing in app
Latency path clientserverDB	PARTIAL	Also clientstorage and edge functions
Bandwidth awareness	PARTIAL	50 MB upload cap; counter: full-table list fetches

System gaps

- Not a single vertical box: Vercel + InsForge + edge functions.
- Multiple deploy units: UI, BaaS schema/functions, cron schedules.
- Shared multi-tenant DB, not one database instance per company.
- No documented 1–30 users per company assumption.
- List fast-load incomplete: unbounded allPages + client pagination.
- In-memory IP rate limit breaks clean load-balancer semantics.

4. Data layer checklist

Agent: Data schema audit. Evidence from ~48 SQL migrations, SDK clients, report RPCs, and entity types. Not UI-only: posting engine, write RPCs, RLS, and reports are in Postgres.

Data stack

Layer	Tech
DB	InsForge Postgres (https://yfmw4i43.eu-central.insforge.app)
Access	@insforge/sdk (no Prisma/Drizzle)
Schema	SQL migrations under migrations/
Files	attachments.key + url object storage
Tenancy	company_id + RLS; one company per user (company_members)

Database and schema rules

Item	Status	Evidence
What to store (accounts, vendors, invoices, stock, sales, users, roles)	PASS	Full ERP tables; vendor = suppliers
Fixed tables	PASS	Explicit CREATE TABLE + company_table_manifest
Relationships / join tables	PARTIAL	Strong FKs; roles = company_members.roles text[] (no usersroles join)
Transactions (atomic multi-write)	PASS	Posting RPCs complete in one PostgreSQL transaction
ACID	PASS	InsForge PostgreSQL
Building blocks (Account, Vendor, Debit, Credit, Balance, Date)	PASS	Balance computed in trial balance RPC
File link in DB only	PASS	attachments.key + url; buckets documents/imports
User access patterns	PASS	RLS + has_company_role / assert_write_capability
App questions (TB, GL, stock, sales)	PARTIAL	TB + P&L/BS/CF; no dedicated GL report; TB unfiltered
Role model	PASS	admin, buyer, warehouse, sales_rep, accountant, ...

Roles and access patterns

Role / block	Status	Note
Accountant (TB, GL, filters, D/C/B)	PARTIAL	TB + statements exist; no dedicated GL; TB has no date/account/vendor filters
Purchasing (POs, vendor buys)	PASS	P2P tables + write RPCs; buyer role
Inventory / data entry	PASS	stock_moves, transfers, adjustments; warehouse role
Sales (docs + customer lookup)	PASS	Q2C tables + customers; sales_rep / ar_clerk
CEO / admin	PASS	invite_user, set_member_roles, settings, summaries
Trial-balance blocks present	PASS	Account / suppliers / debit / credit / balance / date

Data gaps

- No table named vendors — use suppliers (+ vendor_* documents).
- No usersroles join table — roles are company_members.roles text[].
- No stored balances entity — balance is computed in trial balance.
- No dedicated general-ledger report RPC.
- report_trial_balance has no date / account / vendor parameters.
- Role names differ from design labels (buyer vs Purchasing, warehouse vs Inventory).

5. Entity inventory (summary)

Tenancy: companies, user_profiles, company_members, invitations, platform_admins. Master: currencies, branches, warehouses, accounts, products, customers, suppliers, tax_codes, fiscal_periods, ... P2P: purchase orders, goods receipts, vendor bills/payments/returns, RFQs. Q2C: quotes, sales orders, delivery notes, customer invoices/receipts/returns. GL / inventory: journal_entries, journal_entry_lines (debit/credit), stock_moves, adjustments, lots. Ops: bank statements, reconciliation, period close, fixed assets, attachments, write_commands, approvals.

6. Recommended next steps

- A. Rewrite the design document so it matches the real stack (Vercel + InsForge multi-tenant Postgres).
- B. Close concrete gaps: trial-balance filters, dedicated GL report, list paging (stop unbounded allPages), durable rate-limit store.

Note: Official STE compliance needs the ASD dictionary. Source design text followed simple-english / STE rules. This PDF is an audit artifact for atmata-ui-only, not a product specification.